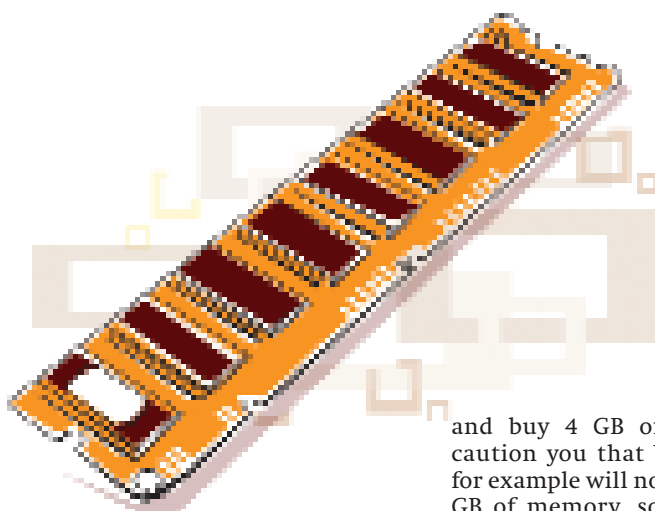




Memory

We all know its significance, but do we all know it's dirt cheap now? Go on, your PC could always use more...

Memory is the most-often upgraded part of any PC. The reason is usually that a huge increment in performance can be gained by just upgrading memory rather than changing a processor or swapping a hard drive. This is because memory is the expressway via which your processor talks to your graphics solution and Northbridge—since there's huge amount of data that flows back and forth from these processing centres (both CPU and GPU), your memory had better be up to the task. If you visualise memory as an expressway then it's easy to understand the concepts of its speed (i.e. frequency) and the amount of it (megabytes, gigabytes etc). In short the faster the memory the better, but what really gives a significant boost to performance is more memory. Just upgrade from 512 MB to 1 GB under Windows XP and you'll notice the difference.

There has always been a sweet spot for the amount of memory needed. It used to be 512 MB a couple of years ago. These days multimedia being the memory hog, it is 2 GB. Some games and video rendering / editing work will even scale up in performance at 3 and 4 gigabytes of memory. Does this sound a costly proposition? At Rs 2,600 for 2 GB, memory is no longer a wallet-busting proposition. We feel the prices of DDR2 memory fill fall further. It's easy to get carried away

and buy 4 GB of memory but we caution you that Windows XP 32-bit for example will not use more than 3.2 GB of memory, so if you must have more memory go 64-bit for your OS, else stick with 3 GB.

Talking DDR2, it's mainly available in two flavours—667 MHz and 800 MHz (clock speeds). The difference in performance in memory-hungry applications will justify the premium the faster memory commands. DDR is officially out of production and supply—and will be costly due to dwindling supplies. DDR3 promises to become the new standard in another year or so, and scales above DDR2 in terms of speed—we've 1333 MHz DDR3 samples, with up to 1800 MHz being promised soon. The only issue with DDR3 right now is the relatively higher latencies which detract from performance—clock per clock, DDR2 is still faster.

All About Timing!

Will all DDR2 800 MHz memory be equally fast? No. Will 1066 MHz memory necessarily be faster than 800 MHz memory? No again. It all depends on the latency of the memory. Latency, as you know, is defined as *wasted clock cycles*, and memory latencies read something like 4-4-4-12 at a particular frequency. Latency and clock speeds are directly proportional. Of course, higher latencies are slower, so the relationship is inversely proportional.

Is All Memory Created Equal?

The sad fact is that most vendors do not know much about memory (other than the density of the stick, and the brand). Even memory chips have premium brands and lower brands—this is aside from the brand. In fact the big memory chip manufacturers are Micron, Samsung, Infineon, Quimonda, Hyundai, and Kingston—most of these brands don't actually market memory, and aside from the latter two, they're solely OEM players. But keep in mind

that there are differences in performance, overclocking capabilities, and stability between different chips of memory even if their clock speeds and timings are the same.

Hoard Up Or Wait?

Intel has announced support for DDR3 as a memory platform due to the increased bandwidth it provides (eventually, DDR3 will scale up to 2 GHz). However DDR2 is as strong as ever, with prices at an all time low. We expect DDR3 to become mainstream in late



2008. For one, the astronomical prices, and secondly, there aren't any significant performance gains as yet. We say upgrade to 2 GB of DDR2 memory to ensure your PC runs everything properly, but don't spend more than 5,000 bucks on memory right now. We estimate DDR3 will become really affordable by early 2009.

What Makes The Best Buy In Terms Of Speed And Quantity?

In terms of speed, DDR2 800 MHz hits the sweet spot between price and performance. Priced at around 600 bucks more for a 2 GB kit, DDR2 800 is a more viable option than the older DDR2 667. Look for latencies of 3-4-4-8 for 667 MHz memory, and 4-4-4-12 for 800 MHz memory. Just make sure to buy a kit, or to buy your 2 GB together (same brand and speed) so that you can use them in dual-channel mode—which makes a difference of 5-12 percent depending on the application. ■



Agent Tips

Don't buy unbranded or locally-branded memory. It's a very stressed-out component in your PC, and the number one cause of freeze-ups and restarts. Invest wisely—a few hundred bucks more for peace of mind is worth it.